

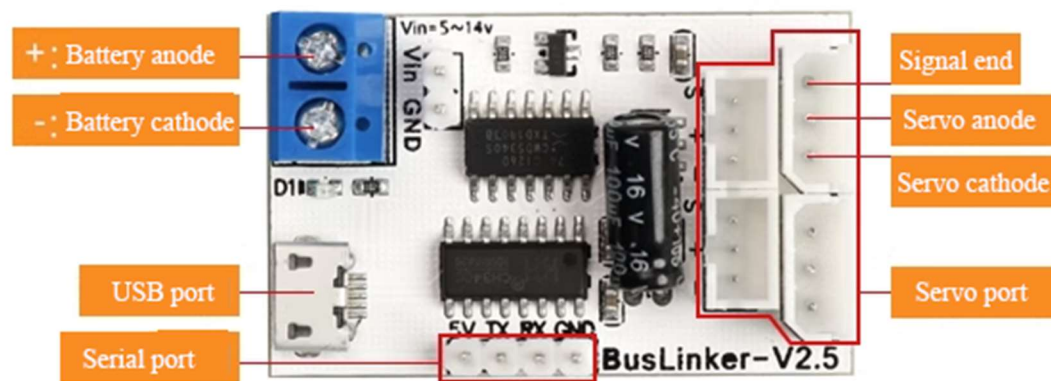
Servo Debug Board User Manual



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Prior to the use of servo, it is necessary to set servo ID in servo debugging software. TTL/USB debugging board is a tool for debugging servos. With the provided Bus Servo Terminal software, it can test servos and set servo parameters. Additionally, the debugging board can communicate via the serial port with a microcontroller to control the servos.



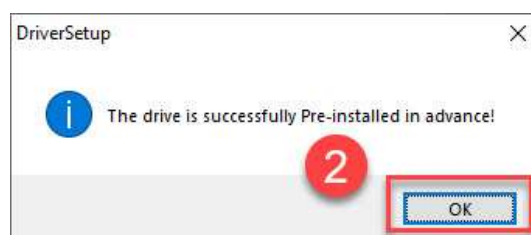
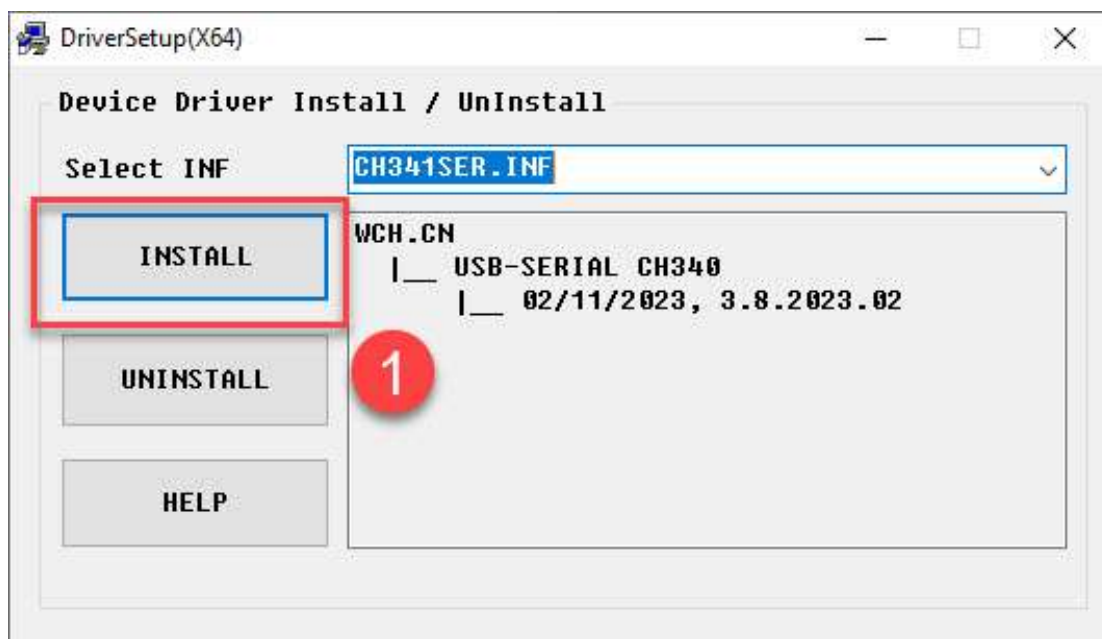
This section explains how to connect a servo to the TTL/USB debugging board and use the Bus Servo Terminal software for servo debugging.

1. Install BusLinker Debug Board Drive

- (1) Locate the “ch341ser.exe” driving package in “08 Servo Debug Board Software/ BusLinker Debug Board Drive”.



- (2) Install the software according to the installation prompts.

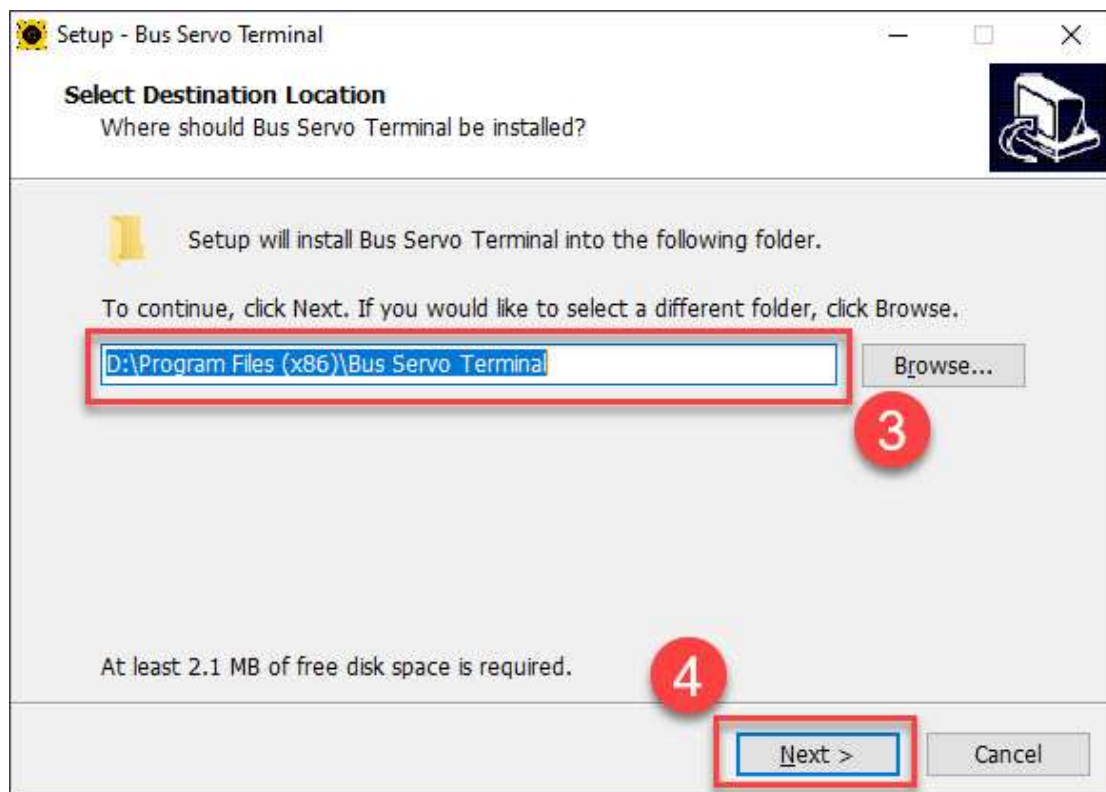


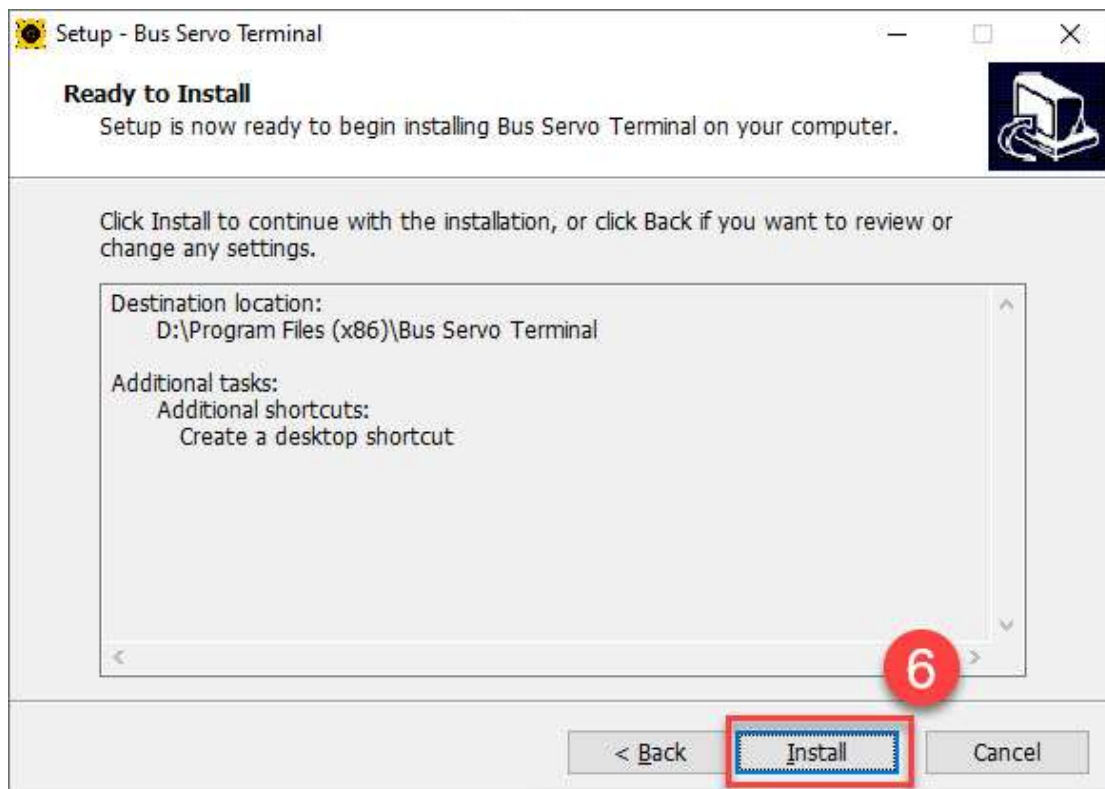
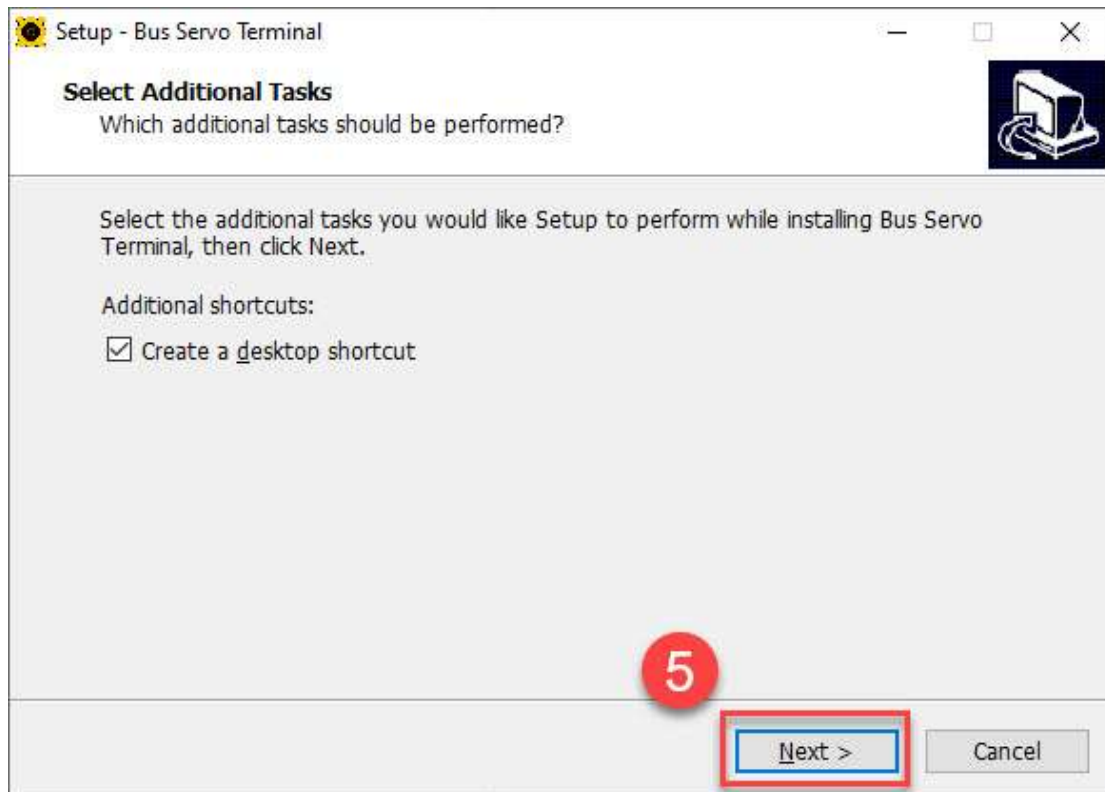
2. Install BusLinker Debug Board Software

After the driving software is install, locate the “**Bus Servo Terminal setup V2.3.exe**” in “**08 Servo Debug Board Software/ BusLinker Debug Board Software**”.



Install the software according to the installation prompts.

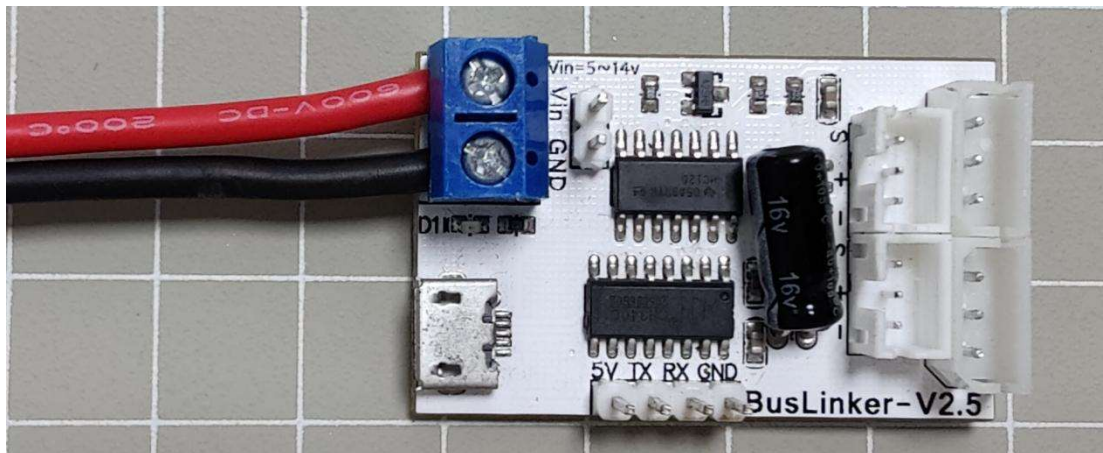






3. Connect Debug Board to Debug Software

- 1) Take out the power cord that comes with the battery. Connect the red wire to the positive terminal (Vin) of the servo debug board, and the black wire to the negative terminal (GND). **Please note that do not directly connect the battery terminals to the debug board before the power cord is connected to the board to avoid short-circuiting the positive and negative terminals.**



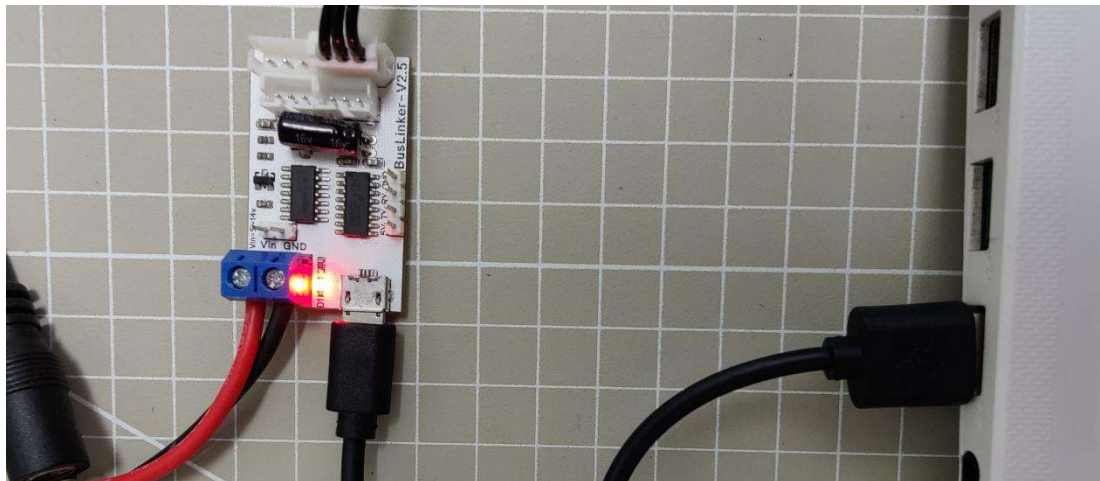
2) Connect the servo to the debug board using a 3-pin cable. The servo interface is designed to prevent reverse insertion. Do not force the cable. See the figure below:



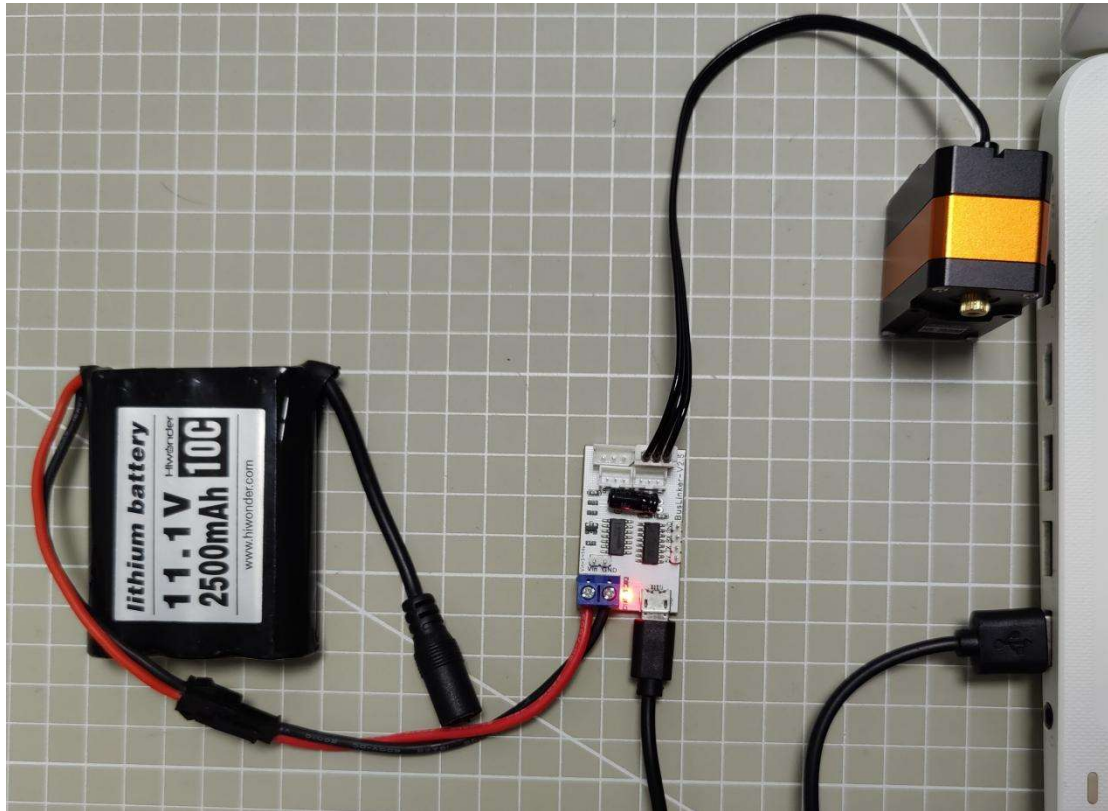
3) After completing the connection, connect the battery terminals with red to red and black to black. **The design prevents reverse insertion, so if it doesn't connect easily, do not force it and change another side.**



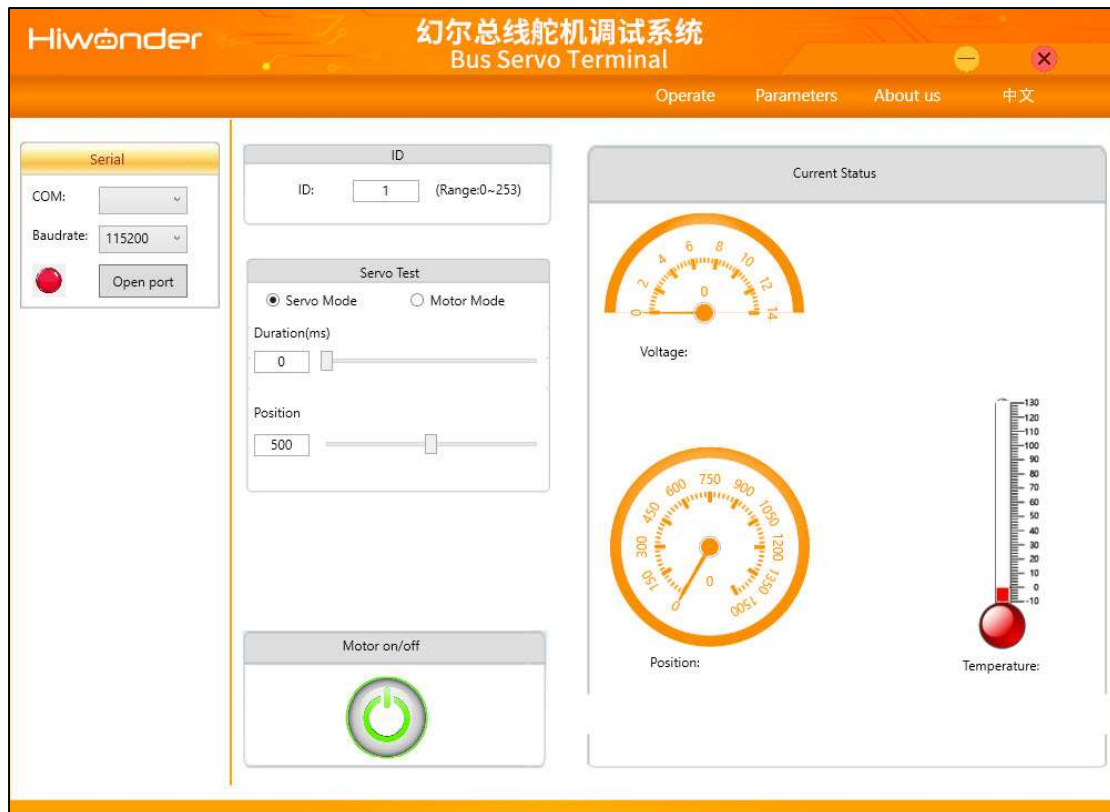
4) Connect the debug board to the PC using a USB cable.



5) The complete wiring diagram is as follow:



6) Double click  to open the servo debug tool.



- 7) Click on the COM port in the serial communication toolbar, and select the port number to connect to. The port number appears randomly, in this example, it is COM3. Do not connect if COM1 appears, as it is the system's communication interface. Here, the "COM3" port is taken as an example.

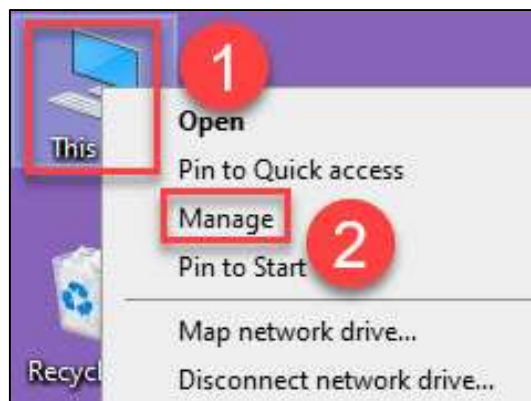


- 8) Now, the servo debug board has successfully established a connection with the servo debug tool.



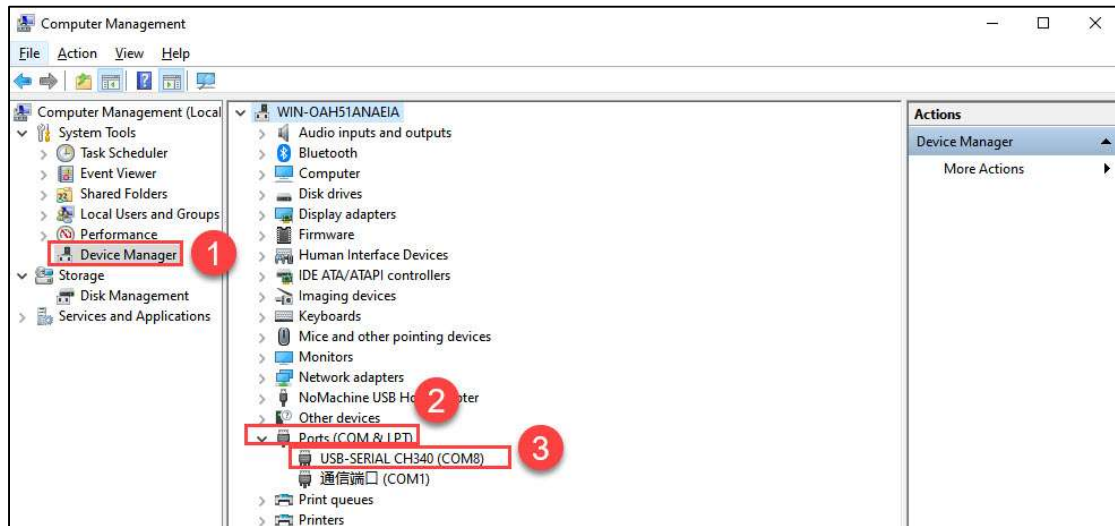
Note: If multiple devices are connected to the computer, you can check the port number of the servo debugging board in the Device Manager. Look for the entry labeled "USB-SERIAL CH340" which indicates our device. The specific steps are as follows:

Right-click on **"This PC"** and click **"Manage"**.



In the Computer Management window that pops up, click on **"Device"**

Manager," then click on the drop-down button for "Ports." The "COM3" port next to "USB-SERIAL CH340" is the port number for our device. The port number appears randomly.



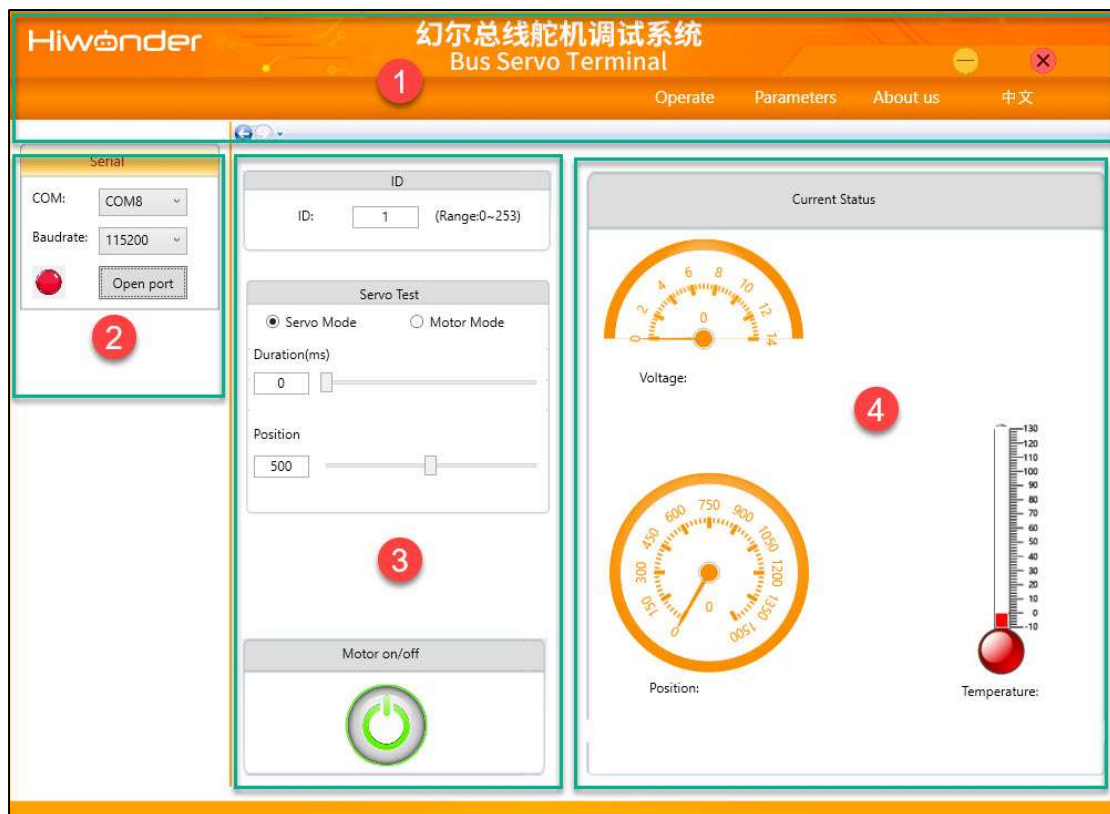
4. Servo Debug Tool Interface Layout

In the servo debug tool interface, there are three main operation interfaces: Basic Operations, Parameters, and About Us. Details about each interface are explained in the following sections.



4.1 Basic Operations

The Basic Operations interface allows real-time operation of the servo and observation of its operational status. Functionally, it can be divided into four parts: the menu bar, debug board connection area, servo control area, and servo status display area, as shown in the diagram below:



- ① Status Bar: close or minimize the software, switch the interface language, and switch between pages. Detailed functionality descriptions can be found in the table below.

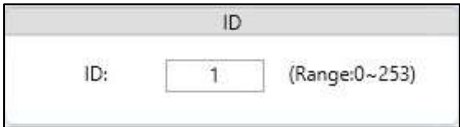
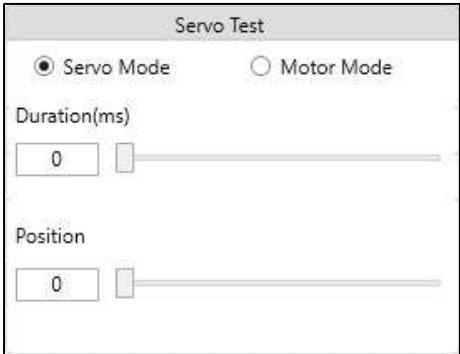
Icon	Function Instruction
	Minimize the interface
	Close the interface
	Switch to servo operation interface
	Switch to servo parameter settings interface

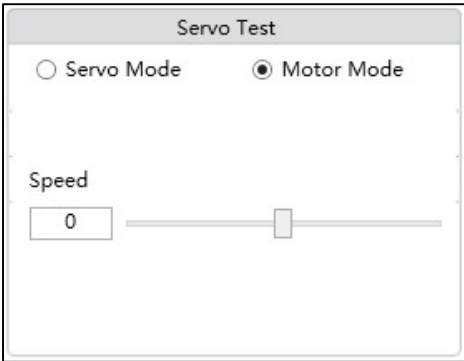
		Hiwonder official information interface
		Set interface language.
		Switch to the next interface from the right side.
		Switch to the next interface from the left side.
		Select and switch to the corresponding interface.

- ② **Debug Board Connection Area:** In this area, you can establish a connection between the debug board and the software. Detailed functionality descriptions can be found in the table below.

Icon	Function Instruction
	Click to select the device port number.
	Click to select the serial baud rate. The baud rate for debug board is 115200.
	Click " Open Port ", the button turns green. Clicking again will close it, and the button turns red.

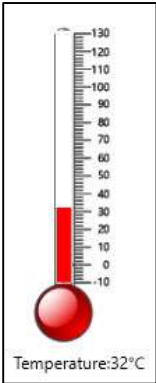
- ③ **Servo Control Area:** Controls the rotation of the servo. Detailed functionality descriptions can be found in the table below.

Icon	Function Instruction
	Input servo ID number to control the corresponding servo.
	Switch to servo mode.
	The time for the servo to move from the current angle to the specified angle, 0~3000ms.
	Servo's rotational position ranges from 0 to 1000, corresponding to angles from 0° to 360°.
	Click to turn off the motor, and change the button color to red. Click it again to turn the motor on and change the button color to green.

		Slide the slider to adjust the speed of the servo's rotation. The greater the absolute value of the speed, the faster the servo rotates. When a negative value is selected, the servo rotates counterclockwise. When a positive value is selected, it rotates clockwise. A value of 0 stops the servo rotation.
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- ④ Status Display area: display the voltage, position and temperature information of the current servo. Detailed function description refers to the below table.

图标	功能说明
	Display the current voltage of servo.

	Display the current position information of the servo.
	Display the current temperature information of the servo.

4.2 Parameters Interface

The parameter setting interface allows for the configuration of basic parameters for the servo. Functionally, it can be divided into three parts: the menu bar, the debug board connection area, and the servo parameter setting area, as shown in the following figure:

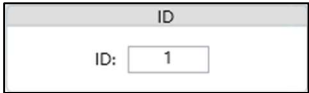
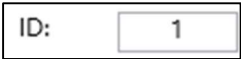
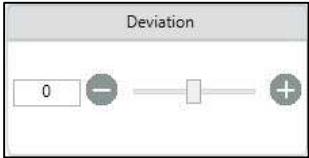


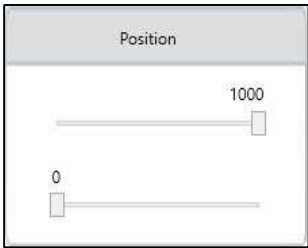
① Status Bar: Close or minimize this software. Switch the language and page of the interface. For detailed functionality description, refer to section 3.4.1 for the contents of the status bar.

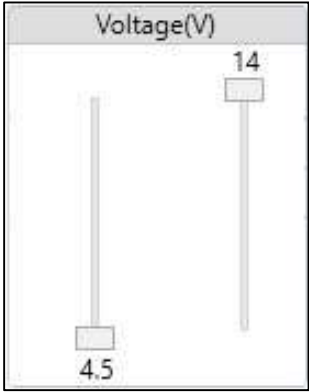
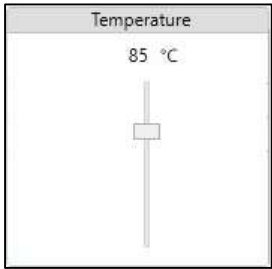
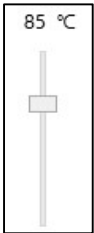
② Debug Board Connection Area: In this area, you can establish a connection between the debug board and the software. For detailed functionality description, refer to section 3.4.1 above for the contents of the debug board connection area.

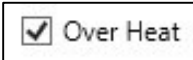
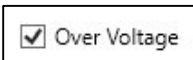
③ Servo Parameter Setting Area: This section allows you to read the servo parameters, set the servo parameters, and restore the servo to its default settings. Specific functionality is described in the table below:

Icon	Function Instruction
	When performing parameter

		settings for the servo, first click the " Read " button to retrieve the current parameter values of the servo motor.
		After completing the modification of servo parameters, click the " Set " button to apply the changes and make them effective.
		Restore the servo parameters to their initial values. Then, click " Default " to apply these initial values to the servo settings.
		The ID number of the servo is typically default set to 1. You can input a number in the text box to set the servo motor's ID, within the range of 0 to 253.
		Adjust the servo offset by dragging the slider, clicking the "+" or "-" icons, or entering a numerical value in the text box. The adjustment range is -125 to 125, corresponding to an angle deviation of $\pm 30^\circ$.

		The adjustment range for 360° magnetic encoder bus servo positions is from 0 to 1500. For other bus servos, the angle range is from 0 to 1000. Adjusting by two sliders. Moving the upper slider to adjust the maximum rotation position of the servo. When the slider is set to 1500, the servo can rotate up to 360°. The angle conversion is calculated by the formula: $360/1500 = 0.24^\circ$.
		Dragging the lower slider to adjust the servo's minimum rotation position. When the slider is at 0, the servo can rotate to a minimum angle of 0°.
		Move the slider to adjust the minimum output voltage, which ranges from 4.5V to 13.9V. After adjustment, if the voltage drops below the minimum value during use, the servo will stop

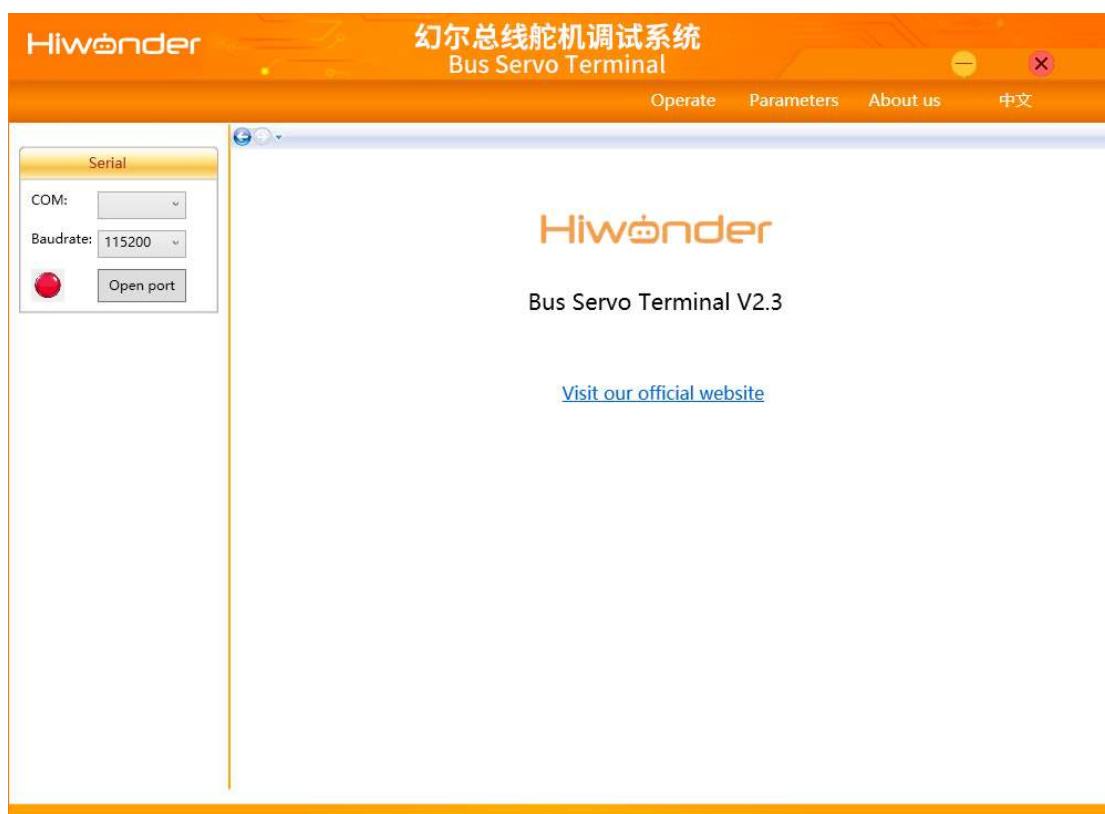
		operating. Drag the slider to adjust the maximum output voltage, within the range of 6.5V to 14V. After setting, if the voltage exceeds the maximum value during use, the servo will trigger an over-voltage alarm.
		Temperature Range Adjustment: Users can set the operating temperature range for the servo. Dragging the slider to adjust within the range of 50 to 100. If the servo exceeds the set temperature during operation, it will trigger a high-temperature alarm. At this point, the servo stops rotating, and its temperature increases rapidly. Immediate power disconnection is necessary, allowing the servo to cool down before resuming operation.

	 	LED Light Control: control the on/off state of the LED light on the servo. Click the icon to toggle the LED light. When the icon is yellow, the LED is in the on state, and when it is gray, the LED is in the off state.
		Over-temperature: When the servo temperature exceeds the set temperature range, the LED light will flash red as an alarm. At this point, immediately disconnect the power and wait for the servo to cool down before resuming use.
		When the servo voltage exceeds the set voltage range, the LED light will flash red as an alarm. At this point, immediately disconnect the power and inspect the circuit for any issues.
		Check Stall: When the servo is stalled, the LED light will flash

		red as an alarm. At this point, immediately disconnect the power and adjust the servo's rotation position.
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4.3 “About Us” Interface

In the “About Us” interface, users can view our official brand website.



5. Usage of the Servo Debug Tool Software

For basic usage of the servo debug tool software, you can refer to the “Bus Servo Debug Board Manual” video located in the same path as this document for learning.